

Nexus Vision AI

Enterprise computer vision software for manufacturing, retail, and logistics operations.

Company Overview

Nexus Vision AI builds production-grade computer vision software that turns camera and sensor feeds into real-time operational intelligence. Founded in 2019, the company serves enterprise manufacturing, retail, and logistics customers across North America and Europe, with over 400 production deployments processing more than 50 million frames per day. Our platform spans edge inference appliances to a fully managed cloud API, so customers can deploy computer vision wherever their operations require it.

Product Portfolio

NexusScan Edge

On-device computer vision appliance for real-time quality inspection on manufacturing lines.

- Sub-15ms inference latency on-device
- Runs on ARM / NVIDIA Jetson edge hardware
- Fully offline-capable, no cloud dependency
- Auto-retrains from flagged defect samples

NexusVision Cloud Platform

Scalable cloud API for large-scale image and video analytics across multi-site operations.

- REST/gRPC API with SDKs for Python, Java, and Go
- Auto-scaling inference clusters
- Built-in data labeling and active-learning loop
- SOC 2 Type II compliant

NexusGuard Retail

Loss-prevention and shelf-monitoring vision suite built for retail environments.

- Real-time shelf out-of-stock detection
- Integrates with existing IP camera infrastructure
- Configurable alerting via webhook, Slack, or Teams

NexusFlow Industrial

High-throughput defect detection for industrial production lines.

- 300 fps inline inspection
- Sub-pixel defect localization
- Integrates with PLC/SCADA systems for automatic line-stop

Technical Specifications

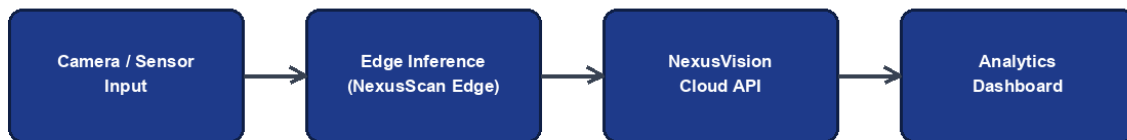
Benchmark figures below are measured on representative production workloads across our current product lines.

Product Line	Deployment Model	Model Accuracy	Inference Latency	Max Throughput
NexusScan Edge	On-Prem Edge Device	98.7%	12 ms	240 fps
NexusVision Cloud Platform	Cloud API	99.2%	45 ms	1,200 req/s
NexusGuard Retail	Hybrid (Edge + Cloud)	97.5%	18 ms	180 fps
NexusFlow Industrial	On-Prem	99.4%	9 ms	300 fps

System Architecture

The diagram below illustrates the standard NexusVision inference pipeline, from camera input through edge inference to cloud aggregation and analytics.

NexusVision AI — Inference Pipeline Architecture



Why Nexus Vision AI

Customers choose Nexus Vision AI to replace manual visual inspection and monitoring workflows with consistent, auditable, real-time computer vision -- deployable at the edge, in the cloud, or both, without vendor lock-in to a single hardware platform.